WORKSHEET

Chapter 1: Equations of planes

1 Which of the following points lie on the plane $2x - 5y + 3z - 8 = 0$?

a $A(4, 2, -3)$

b $B(2, 1, 3)$

c $C(-0.5, -3, -2)$

d $D(-1, 2, 6)$

2 A plane contains the points $A(1, 1, 1)$, $B(0, 1, 2)$ and $C(-1, 1, -1)$.

a Find the normal to the plane.

b Find the vector equation of the plane.

3 A plane contains the point $A(1, -1, 1)$ and the vector $\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ is normal to the plane. Find the vector equation of the plane.

4 A plane contains the point $A(0, 1, 1)$ and the vector $\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ is perpendicular to the plane.

a Find the vector equation of the plane.

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b Find the Cartesian equation of the plane.

c Show that the points $(3, 1, -2)$ and $(-4, 7, -1)$ lie on the plane.

5 A plane passes through $M(1, -1, 3)$ and has the vector $\mathbf{n} = -4\mathbf{i} + 16\mathbf{j} + 4\mathbf{k}$ normal to it.

a Find the vector equation of the plane.

b Find the Cartesian equation of the plane.

6 Find the equation of the plane that contains the point $(1, 0, -1)$ and perpendicular to $\mathbf{n} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

7 Find the equation of the plane through the point $(5, 1, 4)$ and parallel to the plane $x + y - 2z = 0$.

8 Find the equation of the plane passing through $(0, 1, -7)$ and with normal vector $4\mathbf{i} - \mathbf{j} + 6\mathbf{k}$.

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9 Find the equation of the plane that contains the points in each of the following.

a $(0, 1, 1)$, $(1, 1, 0)$ and $(1, 0, 1)$

b $(-1, 1, -1)$, $(0, 1, 1)$ and $(4, -3, 2)$

c $(0, 1, 1)$, $(1, 0, 1)$ and $(1, -3, -1)$

d $(1, -2, 0)$, $(2, 1, 0)$ and $(-1, 1, 2)$

e $(1, 0, -1)$, $(3, 3, 2)$ and $(4, 5, -1)$

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Answers

1 B and C lie on the plane.

2 $\mathbf{n} = -4\mathbf{j}$

b $\mathbf{p} \cdot \mathbf{j} = 1$

3 $x + y - z = -1$

4 **a** $\mathbf{p} \cdot (\mathbf{i} + \mathbf{j} + \mathbf{k}) = 2$

b $x + y + z = 2$

c Both points satisfy the equation of the plane.

5 **a** $\mathbf{p} \cdot (-4\mathbf{i} + 16\mathbf{j} + 4\mathbf{k}) = -8$

b $-4x + 16y + 4z = -8$

6 $2x + 2y - z = 3$

7 $x + y - 2z = -2$

8 $4x - y + 6z = -43$

9 **a** $x + y + z = 2$

b $8x + 7y - 4z = 3$

c $2x + 2y - 3z = -1$

d $6x - 2y + 9z = 10$

e $3x + 9y + z = 2$

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